

PLA Properties

Normal

Mechanical anisotropy

[Orthotropic mechanical properties of PLA materials fabricated by fused deposition modeling](#)

Values assumed for 100% infill

Property	PLA xy	PLA z
Tensile Strength (MPa)	47	30
Elastic Modulus (MPa)	2700	2150
Poissons Ratio	0.43	0.37
Shear strength (MPa)	25.5	22.1
Yield Strength (MPa)	35	20

Modified PLA

[How much STRONGER is PLA+ really?](#)

PolyLite PLA tested as normal **PLA**

PolyMax PLA tested as the modified **PLA +**

PLA + deforms and yields a huge amount before failing, while normal PLA snaps. So under static loading normal PLA is stronger, but under dynamic loading performance, with the modified material.

Impact behaviour is where modified PLA has better performance.

Comparison table

Property	PLA xy	PLA z	PLA+ xy	PLA+ z
UltimateTensile Strength (MPa)	57	43	43	32
3 Point Bending modulus (MPa)	3000	-	2200	-
Hook Static (kg)	72	43	47	34
Impact strength (kJ/m2)	5	-	17	-
Crystallization temperature (C)	60	-	60	-

Bending modulus

In the XY direction

Material	3-point Bending modulus (MPa)
PP	300

Material	3-point Bending modulus (MPa)
Nylon	1000
ABS	1950
PETG	2000
PLA +	2200
PLA	3000
annealed PLA	3200
PETG CF	3900

Light-Weight PLA

Colorfabb Light-Weight PLA in Black

[A 4th dimension for 3D prints](#)

Density can be adjusted by changing the nozzle temperature - the material foams up

Blowing agent - chemical that causes the foaming

Foams up to almost 3 times its volume

Bubbles appear above 215 C

Fully bubbles at 250 C

With bubbling, the extrusion amount needs to be drastically reduced

Stringing increases with increase in foaming

Increase retraction, but be careful not to clog the printer (not more than 2.5mm)

Flow Correction

Temperature (C)	Extrusion Multiplier
200	0.93
210	0.92
220	0.79
230	0.45
240	0.4
250	0.38

Mechanical properties

Tested at 205 (100% density), 225 (67%) and 250 C (44%), and an annealed part at 250 C.

The annealed part was heat treated in a convection oven at 100C for 1h, which led to quite a big deformation of the parts and extra foaming in the z direction. The material properties remained largely unchanged.

Property	205 xy	205 z	225 xy	225 z	250 xy	250 z
Ultimate Tensile Strength (MPa)	48.9	32.3	23.9	13	11.8	4.3

Property	205 xy	205 z	225 xy	225 z	250 xy	250 z
3 Point Bending modulus (MPa)	3200	-	1600	-	2500	-
Hook Static (kg)	58.8	-	29.9	-	17.3	-
Impact strength (kJ/m2)	5.3	2	2.5	2	1.7	1.6

eSun eLW-PLA

[How to Print LW-PLA](#)

Settings

No top or bottom layers, 1 perimeter wall to print in spiral mode

Stringing cannot be eliminated, no matter how much you try, so the part should be designed to have minimum travel.

Parameter	Value
Layer Height	0.25 mm
Initial Layer Line Width	110 %
Perimeter Walls	1
Z -Seam	Exposed
Z -Seam Location	Back
Infill	0%
Temperature	235 C
Flow	55 %
Print Speed	100 mm/s
Wall Speed	40 mm/s
Slow Layers	2
Retract on Layer Change	ON
Retraction Distance	2 mm
Retraction Extra Prime Amount	0.5 mm^3
Fan Speed	20 %
Union Overlapping Volumes	OFF
Surface Mode	Normal
Slicing Tolerance	Middle

N4_LWPLA

For printing with eSun eLW-PLA

Flow rate and dimensions tested with **flowrate_test.stl**

Foamy right after printing.

Waviness can be caused by loose belt

Test	Temperature C	Extrusion Multiplier	Wall Thickness (mm)	XY Size (mm)	Z Size (mm)	Outer Wall Inset
1	235	0.55	1.5	20.5	20	0.1
2	235	0.5	1.8	20.4	20	0.1
3	235	0.45	1.3	20.4	20	0.1
4	235	0.45	1.3	20.1	20	0.3
5	235	0.45	1.0	20	20	0.4

Final settings

Parameter	Value
Layer Height	0.2 mm
Initial Layer Line Width	110 %
Perimeter Walls	1
Z -Seam	Smart
Z -Seam Location	Sharpest
Infill	0%
Temperature	235 C
Flow	45 %
Retract on Layer Change	ON
Retraction Distance	2 mm
Retraction Extra Prime Amount	0.5 mm^3
Fan Speed	20 %
Union Overlapping Volumes	OFF
Surface Mode	Normal
Slicing Tolerance	Middle
Z-Hop when retracted	OFF